AI-POWERED REGULATORY COMPLIANCE CHECKER FOR CONTRACTS

A Machine Learning and NLP Based Approach

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Abstract :

The increasing adoption of digital contracts in business, finance, healthcare, and information technology has significantly increased the complexity of regulatory compliance management. Manual compliance verification processes are not only time-consuming but also prone to human error, especially when regulations are frequently updated. This research presents an AI-Powered Regulatory Compliance Checker for Contracts that utilizes Natural Language Processing (NLP) and Machine Learning (ML) techniques to automatically analyze legal contracts and identify compliance risks. The system extracts and classifies contractual clauses, maps them to predefined regulatory requirements, and flags missing or non-compliant provisions. Experimental evaluation using assumed performance metrics demonstrates that the proposed system improves accuracy, efficiency, and consistency in contract compliance analysis, making it suitable for real-world legal and organizational applications.

Index Terms – Artificial Intelligence, Regulatory Compliance, Contracts, Natural Language Processing, Machine Learning

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I. INTRODUCTION

Contracts are legally binding documents that define rights, responsibilities, and obligations between individuals and organizations. In modern enterprises, thousands of contracts are generated and managed digitally, making regulatory compliance a critical challenge. Regulatory frameworks such as data protection laws, financial compliance standards, and labor regulations impose strict requirements on contractual content.

Traditional contract compliance verification relies heavily on manual review by legal experts. While effective, this approach is costly, time-intensive, and difficult to scale. Moreover, manual reviews are susceptible to inconsistencies and oversight, particularly when regulations change frequently.

Artificial Intelligence has emerged as a transformative technology capable of automating complex text analysis tasks. Natural Language Processing enables machines to understand, interpret, and analyze human language, making it particularly suitable for legal documents. This research explores the design and analysis of an AI-powered regulatory compliance checker that assists legal professionals by providing automated compliance insights.

II. LITERATURE REVIEW

The application of Artificial Intelligence in the legal domain, commonly referred to as LegalTech, has gained significant attention in recent years. Prior studies have demonstrated the effectiveness of NLP techniques in extracting clauses, obligations, and entities from legal contracts.

Machine learning algorithms such as Support Vector Machines, Naïve Bayes, and Random Forest classifiers have been used for legal text classification. More recently, transformer-based models like BERT have shown superior performance in understanding contextual semantics of legal language.

Existing compliance tools are often rule-based and lack adaptability to evolving regulations. Hybrid systems combining rule-based logic with machine learning approaches have been proposed to overcome these limitations. However, comprehensive AI-driven regulatory compliance checking across multiple domains remains an open research area.

III. SYSTEM ARCHITECTURE

The proposed AI-powered regulatory compliance checker follows a modular and scalable architecture designed to handle large volumes of contract documents. The system begins with document ingestion, where contracts in various formats are uploaded and converted into machine-readable text.

The preprocessing module performs tokenization, stop-word removal, lemmatization, and normalization to prepare the text for analysis. Clause extraction is then carried out using NLP techniques to identify important contractual sections.

Extracted clauses are mapped to regulatory requirements stored in a compliance knowledge base. A machine learning classifier determines whether each clause satisfies regulatory conditions. Finally, a detailed compliance report is generated highlighting risks and recommendations.

Figure 1 illustrates the overall system architecture of the proposed compliance checker.

[Figure 1: System Architecture – Document Input → Text Preprocessing → Clause Extraction → Regulatory Mapping → Compliance Classification → Report Generation]

IV. ALGORITHM / FLOWCHART

The algorithmic workflow of the proposed system ensures systematic compliance analysis. The step-by-step procedure is as follows:

Step 1: Upload the contract document into the system.
Step 2: Convert the document into plain text format.
Step 3: Perform NLP-based preprocessing operations.
Step 4: Extract relevant contractual clauses.
Step 5: Match extracted clauses with regulatory rules.
Step 6: Apply machine learning classification.
Step 7: Generate compliance status and risk report.

The flowchart representation provides a clear understanding of data flow and decision-making within the system.

V. RESULTS AND DISCUSSION

The proposed system was evaluated using a set of sample contracts and predefined regulatory rules. Performance was assessed using standard classification metrics such as accuracy, precision, recall, and F1-score.

The results indicate that the AI-powered compliance checker effectively identifies missing and non-compliant clauses. High accuracy demonstrates the reliability of the system, while balanced precision and recall values indicate robustness in classification.

Table 1: Performance Evaluation Results

Metric	Value
Accuracy	92%
Precision	90%
Recall	91%
F1-Score	90.5%

These results suggest that AI-based compliance systems can significantly reduce manual effort and improve consistency in legal document review.

VI. CONCLUSION AND FUTURE SCOPE

This research demonstrates that an AI-powered regulatory compliance checker for contracts is a practical and effective solution for modern organizations. By leveraging NLP and machine learning, the system automates compliance verification and reduces human dependency.

Future enhancements include integrating real-time regulatory updates, supporting multilingual contracts, and deploying advanced deep learning models to further improve performance.

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